

# Zhuo (Jessica) Lyu

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## Education

Ph.D. Economics, University of California, Davis (**UC Davis**). September 2021 - June 2027 (expected).

Research Fields: Energy and Environmental Economics (EEE), Applied Microeconomics, Economic History.

Graduate Exchange Program, UC Berkeley Department of Agricultural and Resource Economics, January-May 2023. Course: ARE 264 Empirical EEE.

M.A. Public Policy with Certificate in Research Methods, University of Chicago (**UChicago**). Graduate with Honors. September 2019 - December 2020.

B.A. Economics, Honors Bachelor's Degree, University of International Business and Economics (**UIBE**). September 2015 - June 2019.

## PhD Dissertation Committee

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## Research

### **Electricity Transmission Lines and Land Development in the Pacific Northwest (Job Market Paper)**

**Abstract:** Multiple national-scale electricity transmission studies have pointed out that expanding transmission significantly is the most cost-effective way to serve the rapidly increasing electricity demand in the long run while supporting more clean energy resources to generate electricity. Yet, it is extremely difficult to build a new transmission line. Landowner resistance to granting a transmission line right-of-way (ROW) can slow down or kill a transmission project. In this paper, I study the build out of electricity transmission lines in the Pacific Northwest by Bonneville Power Administration from 1938 to the early 1940s. I provide the first-of-its-kind estimate on the impact of ROW resulted from building a new transmission line after 60 years up to the present day. Combining public records of ROW tracts, newly digitized historical and contemporary property assessment records, and newly geo-referenced parcel ownership maps, I find that having a ROW on a parcel reduces parcel value per acre by 14.02% to 26.38% relative to what it would have been if absent a ROW. The impact varies across lines, driven mainly by lines' locations and whether the ROWs were acquired through an easement or title ownership. As historical rural lands have transformed into today's suburbs, easement provides opportunities to develop the ROW lands, while a line's location influences the types of land use that develop after the line is built. The implied annual rate of return that could fully compensate for the long-run losses in parcel value ranges from 7.3% to 10.7%, which suggests that at the time of ROW acquisition, there was considerable uncertainty regarding possible future returns from supporting transmission infrastructure. The wealth impact could be a net positive depending on how compensation monies were invested.

**Effects of Charging and Discharging of Battery Storage on California's Wholesale Electricity Prices**  
(*under review*)

**Abstract:** The total battery capacity on California's grid increased tenfold from 2021 to 2023. Estimating the price effects of batteries' charging and discharging is crucial for understanding the effects of battery storage on the wholesale electricity market. In this paper, I regress hourly real-time electricity prices on the daily total battery charging over the charging hours, or the daily total battery discharging over the discharging hours with a panel fixed effects IV model. I find that when batteries discharge one more MWh over the discharging hours, the price in Hour 20 (7pm-8pm) decreases by 0.37 cents/MWh, which equals \$23.87 when multiplying by the average amount of discharging over the three years. No effects are found in hours other than Hour 20. In addition, price effects vary by years, and exist in different hours in different years. The price decrease in Hour 20 can be huge if only looking at 2023, which is 1.2 cents/MWh, or \$131.46. I also discuss where batteries' energy could come from. I discover a competition between on-peak natural gas generators (e.g., combustion turbines and steam turbines) and batteries' discharging in some of the evening peak hours (e.g., Hour 20 and 21), where the generation from these gas generators would go down as batteries increase discharging. The total generation from natural gas, however, has a positive association with batteries' discharging, reflecting that combined-cycle gas generators, not for on-peak, have increasingly been used as part of the base load fleet to meet demand.

## *Work in Progress*

### **Transmission Outages, Wholesale Electricity Prices, and Interregional Transfer Capability**

**Abstract:** Transmission capability on an electric grid can change constantly, affecting electricity generation and consumption while causing price differences across places. In this paper, I estimate the costs of reducing transfer capability between regions using auto transmission outages from 2016 to 2022 on the Malin intertie, which is used to transfer power between California and the Pacific Northwest. I combine the anonymized TADS data from NERC and CAISO's public transmission outage data and find that the directional price spread from the importing to the exporting region responds fairly quickly among import outages, and increases by \$65.42/MWh 40 minutes after outage starts. For export outages, the responses show up very late, with an average of \$39.3/MWh within 15 minutes after outage has begun for six hours. I then calculate the incremental welfare loss caused by an outage, assuming that the lost in power flows equals the outage-reduced lost in transfer capability. I find no robust evidence of welfare loss from export outages, but the welfare loss from import outages is positive. Comparing with a time-equivalent carrying cost calculated from a CAISO project designed to increase import capability, I show that the import welfare loss is almost half of the equivalent cost, suggesting that lowering outage-related costs alone may not be enough to justify expanding interregional transfer capability.

### **Rethinking Cost Components of Retail Electricity Prices: A Panel Regression Approach**

**Abstract:** Distortions in the retail electricity market bring challenges to set prices equal the short-run social marginal costs of providing electricity. Justifications are often made case by case as to whether we should include the costs of capacity in the short-run private marginal costs. In this paper, I test if it's reasonable to do so with a fixed effects panel regression where I use utility fixed effects to capture capacity costs. Using a panel of utility level electricity prices and costs across US from 2014 to 2016, I find that utility fixed effects don't have statistically significant impact on the short-run pmc, supporting the assumption that the short-run pmc is correctly measured by including only wholesale electricity prices and costs from distribution losses.

### **Storm Recovery and Grid Resiliency (with Niaoniao You)**

## **Research and Professional Experience**

**California Independent System Operator (CAISO)** Summer intern in Market Policy and Development (MPD). June-August 2024.

**UC Davis** Research Assistant (RA) for James Bushnell and Kevin Novan. Fall 2023 and Winter 2024.

Identified the existence of and estimated the magnitude of leakage that resulted from California's cap and trade program using economics regressions.

**UC Davis** RA for James Bushnell. Spring 2023 and Summer 2023.

Built datasets from various U.S. electricity data sources (CEMS, EIA923, EIA860 and EIA930) for a simulation model that investigates how uncertainty affects the implementation of the Inflation Reduction Act (IRA). Conducted high-level literature review over a few electric sector models (e.g., EIA NEMS, EPRI REGEN).

**UC Davis** RA for James Bushnell and David Rapson. Summer 2022.

Made graphs using National Household Travel Survey (NHTS) Data and EIA Wholesale Electricity and Natural Gas Market Data for *The Limits and Costs of Full Electrification* (Rapson and Bushnell, 2024).

**UChicago** RA for Fiona Burlig, Summer 2020; Yana Gallen, Spring 2020; and Eyal Frank, Fall 2020.

**Tsinghua University** Intern at the Center for Energy Transition and Social Development. Spring 2021.

**Peking University** RA at Institute of New Structural Economics. 2018-2019.

## Fellowships and Awards

**UC Davis** Department of Economics

Nonresident Tuition Fellowship, 2021-2026. Research Award, Summer 2025. Travel Award, Spring 2026, Fall 2024, and Fall 2022.

Western Power Trading Forum Fellowship, Winter 2024. (\$10,000)

UChicago DWR Graduate Fellowship at Energy Policy Institute (EPIC), Summer 2020; UChicago Harris School of Public Policy Merit Scholarship, 2019-2020. (\$10,000)

UIBE School of International Trade Outstanding Student Award, 2017-2018.

## Teaching Experience

**UC Davis**, Department of Economics. Teaching assistant (TA) for

Energy Economics (James Bushnell). Spring 2022, Spring 2023.

Intermediate Microeconomics (James Bushnell, Fall 2021, Fall 2022, Fall 2024; Erich Muehlegger, Spring 2025, Spring 2026; Shahar Sansani, Winter 2022).

Transportation and Urban Economics (Erich Muehlegger). Fall 2025.

World Economic History (Christopher Meissner). Winter 2025, Winter 2026.

U.S. Economic History (Christopher Meissner, Spring 2024; Santiago Pérez, Winter 2023).

**UIBE**, School of International Trade and Economics. TA for

Econometrics (Zhihong Chen). Spring 2018.

## Professional Activities

*Presentations* (\*: scheduled)

Southern Economic Association (SEA) 2026 Annual Meeting\*. SEA 2024. Economic History Association (EHA) 2026 Poster Session\*. Bay Area Economic History Workshop, 2025, 2026. Sloan/Berkeley Summer School in EEE, Egg Timer, 2022. UC Davis EEE Group, Economic History Group, Applied Micro Group.

### *Service*

Graduate Student Mentor, UC Davis Department of Economics, 2023-2024. Representative, UChicago Harris Student Government (HSG) Academic Committee, 2019-2020.

### *Other Conferences and Meetings Attended*

UC Berkeley Energy Institute POWER Conference, 2022-2026. Western Power Trading Forum Winter General Meeting, 2024.

### **Others**

Software skills: Python, Stata, ArcGIS, R,  $\LaTeX$ . Language: English (fluent), Chinese (native).

Personal information: Chinese citizen, F-1 student visa.

Last updated: September 15, 2026